- Powder Metallurgy products range from aerospace and automobile parts to engineering and household appliance components.
- Some of the important P/M components include porous parts like bearings and filters, cemented carbide tools and electrical contacts.
- Porous components like bearings, filters and porous electrodes form one of the major area of application for P/M Parts.
- These parts are used for a wide range of industrial applications.

Introduction to Porous Materials

- Porous materials are typical P/M products, which have controlled amount of interconnected porosity in-built into the component for specific applications.
- These engineered components are unique to P/M and are difficult to produce by other techniques.
- Porosity is usually expressed as a percentage and is obtained by subtracting the density (%) of the material from 100.
- The pores represent the open volume in a given material and may be classified as closed and interconnected porosity.
- Closed porosity refers to the amount of isolated or closed pores within the material. These pores do not serve any useful purpose and must be minimized.
- Interconnected porosity, on the other hand, refers to that fraction of porosity, which are connected with one another and to the external surface.
- The amount of porosity will be dependent on shape, size and size distribution of pores.
- The main feature of these products is that the presence of interconnected pores enables flow of fluids (both liquids and gases) to flow through these pores.

P/M Porous Bearings

- Sintered-metal self-lubricating bearings, produced by Powder-Metallurgy route are a unique P/M product and one of the large volume products made by P/M.
- They are economical, suitable for high-production rates and can be manufactured to precision tolerances.
- Sintered-metal self- lubricating bearings are widely used in home appliances (Fans), small motors, machine tools, aircraft and automobiles, business machines, farm and construction equipment.

Important requirements

- Good lubrication characteristics with minimum maintenance and long service life.
- Adequate mechanical strength including fatigue strength over a range of temperatures.
- Good wear characteristics.
- Good heat dissipation capacity.
- Ability to withstand both static and impact loads.
- Must not weld/seize with the mating parts (e.g. shafts) during service.

- An important aspect deciding the performance and service life of the bearing is its design and fabrication.
- Key factors in this respect are the physical properties (thermal conductivity, melting point) as well as the Performance Factor or simply PV factor.
- In other words, the load- carrying capacity of porous-metal bearings is usually measured based on the friction/wear criterion, which signifies the heat generated by the bearing.
- The PV factor is the product of the bearing load, P, expressed in pounds per square inch of projected bearing area, and the surface velocity V of the shaft expressed in feet per minute.
- PV factor of a bearing is specified by a limiting value, which should not be exceeded during service for satisfactory bearing performance.
- For example, in the case of a porous metal bearing, a PV value of 50,000 is usually specified.
- This value is 10,000 for thrust bearings.

- Under service conditions, this design or working (PV) value is much less, since in many cases the sliding velocity is not adequate enough to maintain an effective lubricating film.
- PV limit is also affected by the static load-carrying capacity of the material, which should not be exceeded in service.
- The static load-carrying capacity is dependent on environmental factors, bearing clearances, geometry and the nature of the load (continuous, intermittent or shock loading).
- In case of bearings subjected to continuous operation for prolonged periods or when operating at higher temperatures, the PV factor approaches the limiting value.
- It is possible to replenish the lubricant supply by applying oil at the outer diameter or at the ends of the bearing.
- The oil gets sucked into the pores by capillary action.

- After sintering, the bearing must be sized to the specific dimensions.
- Sizing reduces interconnected porosity and gives greater strength, with lower ductility and a smooth finish.
- Bearings come with or without a steel backing.
- Steel-backed bearings are more common than bearings with no backing material.
- Teflon bearings do not have backing.
- Both Teflon as well as unbacked bearings are employed for specialized applications.

- P/M bearing materials are characterized by multiphase structures.
- These include two-phase alloys, immiscible components like plastics as well as lubricants (liquid or solid).
- P/M bearings may be (i) porous type infiltrated with a lubricant or (ii) dry type containing graphite or Teflon.

- Oil impregnated bearings
- Most porous-metal bearings are made of either bronze or iron, with 10 to 35 vol% interconnected voids or porosity.
- In operation, lubricating oil stored in these pores feed the bearing surface through these interconnected pores, when the temperature increases during operation.
- Any oil, which is forced from the loaded zone of the bearing, is later reabsorbed by the pores through capillary action.
- Since these bearings can operate for long periods of time without additional supply of lubricant, they can be used in inaccessible or inconvenient places where re-lubrication would be difficult.

- It is possible to have variations in the composition to suit the service requirements.
- Addition of 1 to 3.5 wt% graphite can enhance self-lubricating properties.
- High porosity with a maximum amount of lubricating oil is used for high-speed light-load applications, such as fractional-horsepower motor bearings.
- A low-oil-content low- porosity material with high-graphite content is more satisfactory for oscillating and reciprocating motions where it is difficult to build up an oil film.

Features of oil-impregnated bearings

- Excellent self-lubrication properties
- wide range of operating temperature (–200°C to 250°C)
- low-thermal expansion
- no water absorption
- silent operation.

- The extended bearing life, even in severe environments with no need for external manual lubrication, reduced maintenance times and extended service life are other attractive advantages.

Materials used in the manufacture P/M-bearing applications

- Bronze
- Most common porous-bearing material with 90% copper and 10% tin.
- Addition of graphite (1%) is also done.
- These bearings can be produced with a wide range of porosities.
- These bearings have excellent wear resistance, ductility, and corrosion resistance.
- Ease of lubrication, low cost and ease of production gives this material a wide range of applications.

- Leaded bronzes
- Both tin and copper contents are lower in comparison to the bronzes.
- Lead content can go up to 16%.
- The main advantages of this material are its
- lower coefficient of friction
- higher conformability than the 90–10 bronzes
- good resistance to galling, which is advantageous in case the lubricant supply is interrupted.

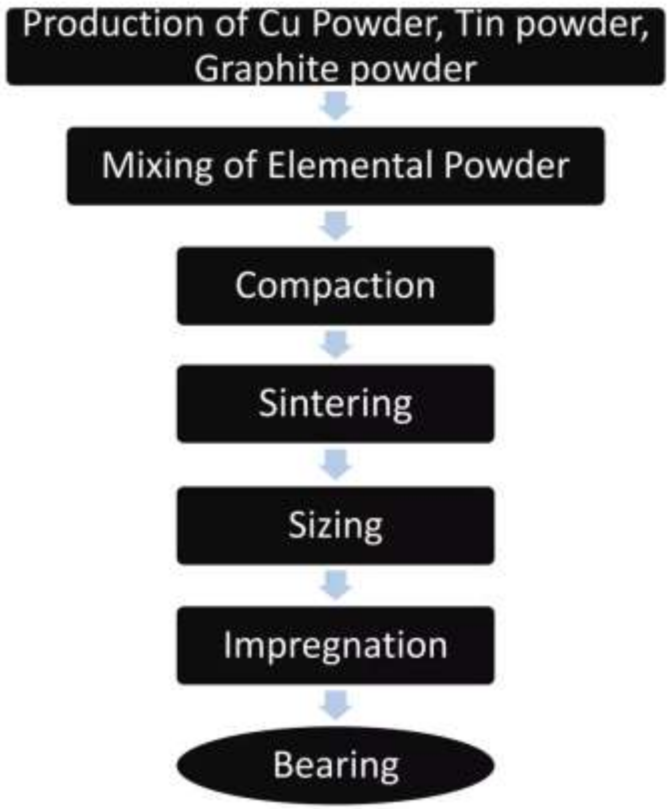
- Iron
- low-cost bearings, widely used in automotive applications farm equipment, and machine tools.
- Copper (up to 10%) is added to iron powders for improving the compressive strength.
- These materials have a relatively low-limiting value of V in the PV factor, but have high-oil-volume capacity because of the high porosity.
- They have good wear resistance and are used with hardened and ground steel shafts for heavy and shock loads.
- Addition of free carbon up to 1.5% to iron-copper makes the material heat treatable.
- In the case of iron-graphite mixture, formation of cementite occurs leading to poor-bearing properties.
- However, the addition of copper (2 to 8%) serves to suppress the formation of cementite and makes iron-copper graphite amenable to heat treatment.
- Heat-treated parts can have a hardness of 65 Rc and also exhibit higher impact resistance for use with hardened and ground shafts.
- Compositions having copper, tin and graphite along with iron are used to make iron-bronze bearings.

- Leaded iron Addition of lead to iron improves speed capability and reduces cost.
- Aluminium Aluminium is used in some applications as they provide cooler operating conditions, light weight, greater tolerance for misalignment and longer oil life than porous bronze or iron.
- The limiting PV value is 50,000, the same as that for porous bronze and porous iron bearings.

Production of Porous Bearings

- Porous bearings account for a majority of the P/M bearings in use.
- These bearings, in addition to the requirements mentioned earlier, must have sufficient amount of interconnected porosity, uniformly distributed throughout the matrix.
- They must also meet the desired strength levels and dimensional accuracy.
- Production of porous bearings, which are subsequently infiltrated with either (i) oil or (ii) plastic.

Steps in the manufacture of porous bearings



- Powder selection: Powders for bronze base bearings use elemental copper powder made by electrolytic, atomized or reduction route. Tin powders are made by atomization. In the case of iron base bearings, atomized or reduced powders are employed.

- Mixing: Mixing of elemental powders is done prior to compaction. For making bronze bearings, copper powder is mixed with tin and graphite powders. Lead powder is added in lead bronzes. Both graphite and lead serve to improve the pressing characteristics as well as to enhance the lubrication capabilities. It is important to ensure thorough mixing in order to avoid segregation of tin particles during sintering resulting in non-uniform sintering.

- Compaction: The mixed powders are compacted to the required dimensions using automated compaction presses.
- Powder is pressed at room temperature.
- Proper green density is achieved by controlling the compaction pressure.
- Pressures used for bronze bearings are in the range of 275 to 480 MPa.
- For iron parts, higher pressures are employed.
- Green density distribution is important factor: uneven green density distribution may cause non- uniform sintering and lead to poor oil impregnation.

- Sintering: Sintering is carried out in continuous mesh-belt furnaces under a reducing atmosphere.
- During sintering, the lubricants are driven off at temperatures between 400 and 450°C.
- During this stage, the low-melting tin powders melt and diffuse into copper matrix.
- Sintering temperature is then raised to 800 to 850°C and the compacts are held for times ranging from 5 to 8 minutes.
- Final microstructure shows a three-phase structure consisting of a homogeneous alpha (copper-tin) phase with graphite particles and pores distributed uniformly in the matrix.
- In the case of iron bearings, sintering process is similar to that of bronze bearings, but sintering is carried out at temperatures of 1,100°C in a protective atmosphere furnace.
- Sintering may cause growth of the compact, which must be taken care of during design stage.

- Sizing: All self-lubricating bearings are 'sized' after sintering to control their dimensions within the allowed tolerances.
- This is to enable the smooth operation of porous- metal bearings, which require suitable clearances between shaft and housing.
- Standard charts are available indicating the recommended bearing clearances for different types of bearings.
- Standards are also available for recommended shaft size and bore diameter to be used with various bearing sizes.

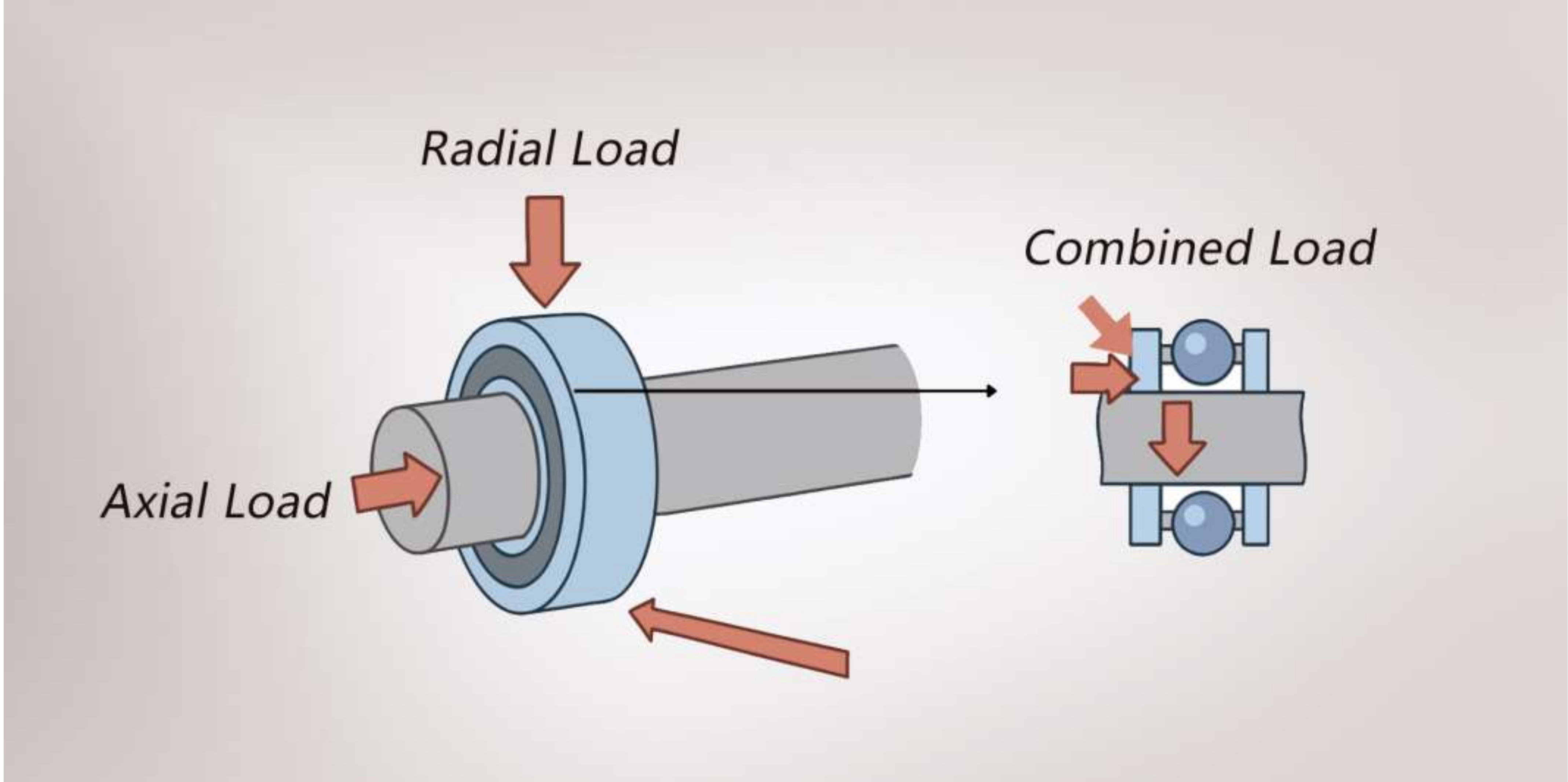
Infiltration

- After sizing, the bearings are impregnated with the lubricant.
- This term refers to the filling of the pores in the sintered part with either oil or a polymer.
- Oil impregnation can be done using immersion in hot oil at 80 to 100°C for 30 minutes.
- Longer impregnation times, oxidation of oil at high temperature are some of the limitations of immersion technique.
- Impregnation is also carried out under vacuum/high pressure or a combination of both.

- The most widely used technique involves a combination of vacuum and pressure.
- In this approach, the sintered parts are placed in a basket and subjected to vacuum (1 mm of mercury or less) to remove all the moisture and air entrapped within the pores.
- The oil is then introduced into the chamber and allowed to enter the pores.
- External pressure may be employed using air or nitrogen at 4 to 5 atmosphere (60 to 70psi) pressure to aid the process.
- The pressure is then slowly brought to the atmospheric pressure and the excess oil is drained.
- This method is quick and efficient than immersion technique.
- Resin impregnation involves the use of polymers such as Teflon or polyacetal to fill up the interconnected pores.
- Resin impregnation improves the density of the parts significantly but renders parts non-heat treatable.

Testing of Bearings

- The bearings are mainly tested for their radial crushing strength of the bearings.
- $P = KLT^2/(D - T)$
- where
- P = radial crushing load (kg)
- K = strength constant for the bearing grade
- L = length of the bearing (cm)
- D = outer diameter of the bearing (cm), and
- T = wall thickness of the bearing (cm)


Radial Load
Combined Load
Axial Load

- Radial crushing strength of a cylindrical bearing is determined by subjecting the bearing to compressive load between two flat surfaces at a no load speed of 2.5 mm/min.
- The load is applied perpendicular to the longitudinal axis of the test specimen.
- The load corresponding to the appearance of first cracks in the specimen is noted.
- The radial crushing strength should exceed the value calculated from the above formula.

Applications of P/M Bearings

- Applications of infiltrated bearings:
- Self-lubricating bearings are used in fractional horse power motors used in electric fans, vacuum cleaners, washing machines, sewing machines, food mixers, refrigerators, air-conditioners, textile and office equipment.
- Their importance lies in the fact that they are very useful where external lubrication is difficult or to be avoided (e.g. food industries)

- Applications of non-porous bearings:
- Copper-lead bearings are employed in automotive applications and for low-duty applications in compressors and gearboxes.
- Heavy- duty application bearings are made from copper-lead alloys having an overlay layer of lead or lead-tin alloy.
- Such an overlay confers excellent fatigue properties to the bearing besides improving anti seizure characteristics and corrosion resistance to oil.